

Bayesian Modeling of Premier League Team Points

Probabilistic forecasting of league tables from match-level data

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1. Problem Formulation

1.1 Problem Statement

English Premier League standings emerge from 380 matches per season (20 teams, double round-robin). At the end of every season, each club's final ranking is the product of thousands of small random events - individual goals, injuries, refereeing decisions - aggregated through a long, noisy season. This makes the league table a natural candidate for probabilistic modelling: instead of predicting a single deterministic outcome, we want a model that expresses genuine uncertainty about a club's final point tally.

We build a probabilistic model that explains end-of-season points for an individual team - including seasons involving promotion and relegation, which is one of the structurally hardest aspects of football forecasting because three clubs enter the league each year with no recent top-flight track record. Our approach works directly on team-season observations: one row per club per season, containing final points and process features derived from match data. This allows us to reason directly about the quantity we care about - the total points - rather than inferring it from thousands of individually simulated matches.

Working at the team-season level also carries a cost that we are explicit about: we lose the ability to model head-to-head dynamics (e.g., how Team A historically performs against Team B) and we cannot simulate mid-season narratives. We accept this trade-off because our objective is the final distribution of points, not the week-by-week storyline.

1.2 Objectives and Use Cases

The project has four concrete, testable objectives, each mapped to a specific notebook and evaluation metric:

- **Backtest:** train on seasons 2009/10-2024/25 ($N = 320$ team-season rows). For each club in 2025/26, predict its season points (`predict_team_points`) and compare to the known final tally. The primary evaluation metric is mean absolute error in points; we also report position MAE after ranking the 20 marginal predictions into an implied league table. This is a sanity check on out-of-sample team-season point forecasts, not on a jointly modelled table. If the model cannot predict points for a season we already know, it should not be trusted for genuine forecasting.
- **Forecast:** predict the 2026/27 table for the confirmed 20-team squad (17 returning clubs + 3 promoted: Coventry, Hull City, Sunderland), producing both a point estimate and a full posterior distribution over final points and position for every club.
- **Team rating:** estimate relative long-run team quality from posterior team-strength parameters. In Model 2, persistent `team_skill` is explicitly separated from shared `season_effect`, which helps answer questions like “is Liverpool genuinely stronger than Chelsea, or did they mainly benefit from a strong season?” In Model 1, long-run strength is captured by a static `skill[team]`, but season-to-season variation is partly absorbed by process/lag covariates rather than a dedicated season-level parameter.
- **Model choice:** compare Model 1 (static latent strength with process/lag covariates) against Model 2 (hierarchical latent strength with explicit season effects and no process covariates), using formal Bayesian comparison on training data (PSIS-LOO, WAIC) and a practical holdout evaluation on 2025/26 - mean absolute error in predicted team points, with position MAE on the implied league table as a derived summary metric.

1.3 Data Source and Description

Source: datahub.io English Premier League CSVs (originally football-data.co.uk, a long-standing, widely used free source for football statistics). We use 17 seasons (season-0910.csv ... season-2526.csv), giving 6,460 individual matches across 42 distinct clubs that have appeared in the Premier League at some point in this window (the 20-club division turns over roughly 3 clubs per season via promotion/relegation, so many more than 20 distinct clubs appear across 17 years).

Key fields used from the raw match-level CSVs: Date, HomeTeam, AwayTeam, full-time goals (FTHG, FTAG), full-time result (FTR: H/D/A), and shots on target (HST, AST). We deliberately avoid betting-odds columns and advanced metrics (xG, possession) that are not consistently available across all 17 seasons - using only fields with full historical coverage avoids introducing systematic missing-data bias in older seasons.

Statistic	Value
Seasons	17 (2009/10 - 2025/26)
Total matches	6,460
Distinct clubs	42
Average goals per match	2.80
Home win rate	45.1%
Away win rate	30.6%
Draw rate	24.3%

The home win rate (45.1%) being roughly $1.5\times$ the away win rate (30.6%) reflects the well-documented home advantage in football - travel fatigue, crowd support, and referee bias are the usual explanations in the sports-analytics literature. This asymmetry is part of the motivation for engineering a shots-on-target differential feature (Section 1.4) rather than relying on raw goal counts, which are noisier and more home-biased.

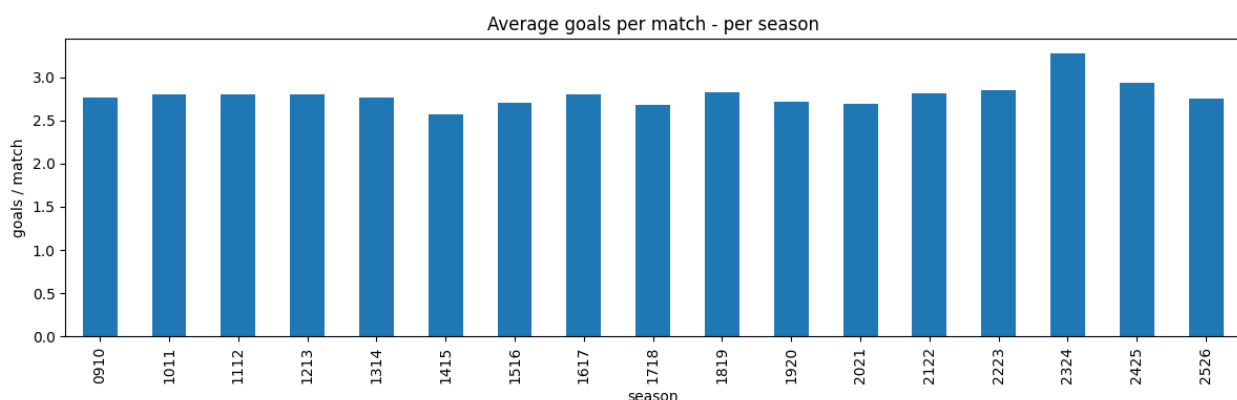


Figure 1.1 - Goals per season across all 17 Premier League seasons.

Figure 1.1 shows that both the average goals per match and the home/away/draw split have stayed remarkably stable across the 17-season window - there is no obvious structural break (e.g. a rule change or VAR-era shift) that would invalidate pooling all seasons into a single training set. This stability is itself a modelling assumption we rely on implicitly: we are treating "team quality" as estimable from a stationary process.

1.4 Preprocessing

Converting 6,460 raw match rows into 340 team-season rows (one per club per season, 2009/10–2025/26) required several deliberate engineering decisions, each aimed at avoiding data leakage and producing features on a comparable scale:

- Load all season CSVs, parse dates, compute total_goals for EDA only.
- Aggregate end-of-season tables via `compute_table()` (points, GD, position; tie-break: Pts → GD → GF).
- Lag features from season s-1 to predict season s (sot_diff_pg, pts_lag1, ppg_last10) - main anti-leakage step.
- Z-score continuous covariates on training data only; apply the same means/SDs to test and forecast.
- Index teams (1...T) and, for Model 2, seasons (1...S) for Stan.
- Promoted teams: `is_promoted` = 1; continuous covariates set to raw 0, then z-scored (below-average, not $z = 0$); penalty mainly via `beta_promoted`.

Model inputs after preprocessing: Model 1 uses z-scored covariates + `is_promoted`; Model 2 uses only team, season, `is_promoted`, and Pts. Backtest trains on 320 rows (through 2024/25); full-data training uses 340 rows. League tables at evaluation are built by ranking per-team point forecasts, not predicted jointly.

Feature	Mean	Std	Min	Max	Description
Pts (target)	52.42	17.70	12	100	End-of-season points
sot_diff_pg	0.30	1.70	-3.76	+6.03	Shots-on-target difference per game (prev season)
pts_lag1	45.02	26.80	0	100	Points in previous season (0 for promoted)
ppg_last10	1.21	0.78	0	3	Points per game in last 10 games (prev season)
is_promoted	-	-	0	1	Binary flag: 1 if newly promoted

1.5 DAG and Confounding Structure

A Directed Acyclic Graph (DAG) makes explicit which variables are assumed to influence which others in the generative model: priors and observed inputs feed into a linear predictor μ , which parameterises the likelihood of observed season points.

This is useful for two reasons. First, the DAG and the Stan equations in Section 2 are two views of the same assumptions. Second, it clarifies confounding: a club's underlying long-run quality influences several observed covariates in Model 1 - strong clubs tend to have high `pts_lag1`, `ppg_last10`, and `sot_diff_pg`. Model 2 addresses this differently: it models persistent `team_skill` and a shared `season_effect` explicitly and does not include the process/lag covariates in the likelihood.

Reading the DAG

- **Model 1:** observed lagged covariates (`pts_lag1`, `ppg_last10`, `sot_diff_pg` from season s), `is_promoted`, and latent static skill[t] (constant across seasons) feed into μ ; then $Pts_{s,t} \sim \text{Student-t}(5, \mu, \sigma)$.
- **Model 2:** only `is_promoted`, latent `team_skill`[t], and latent `season_effect`[s] (a league-wide season shift, shared by all clubs in season s) feed into μ ; same Student- t likelihood. The three process covariates are not parents of μ in Model 2.

In both models, one output row is one team-season: the observed target is that club's season points; a full league table is obtained only later by ranking separate point predictions.

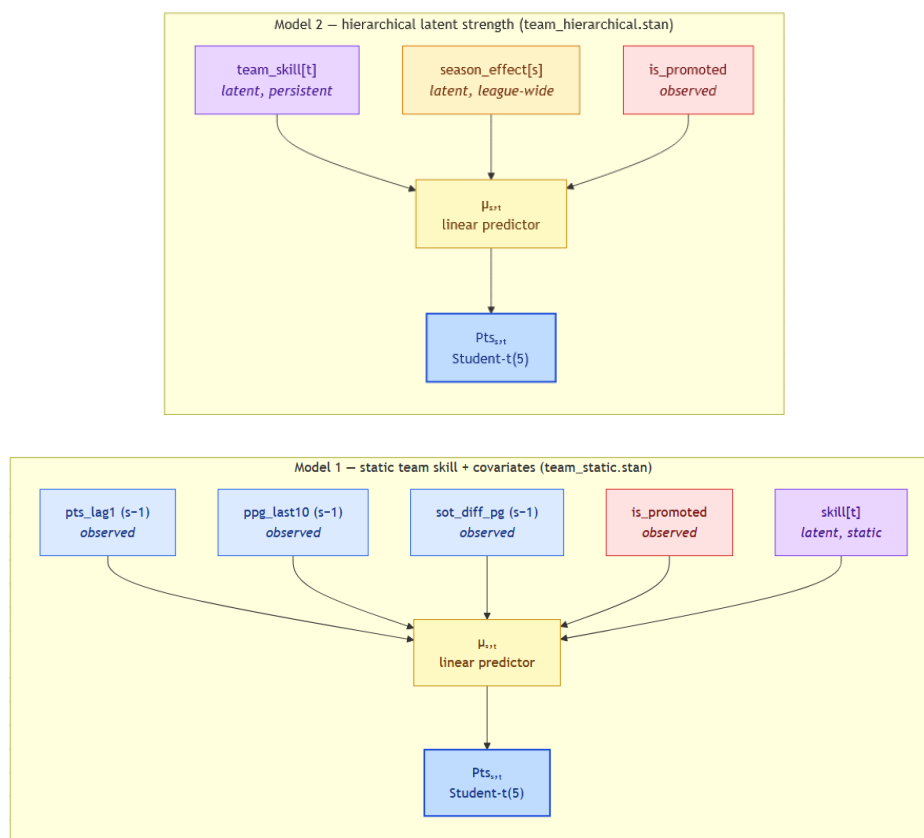


Figure 1.2 - DAG of the generative models.

2. Model Specification

Both models are Bayesian regression models for team-season points, with a Student-t likelihood on the points scale, fit in Stan via CmdStanPy. Each observation is one club in one season, the target is end-of-season points Pts , not a jointly modelled league table.

The comparison is substantive, not a pure “skill-only” ablation: Model 1 adds lagged process covariates to a static latent team strength; Model 2 omits those covariates and instead uses a hierarchical latent structure with persistent team quality and a shared league-wide season effect. Any differences in fit or forecast quality (Sections 4-6) therefore reflect both the skill representation and the choice between covariate-based vs latent-structure explanations of season-to-season variation.

Shared choices: Student-t likelihood with fixed $\nu = 5$ (heavier tails than Normal - extreme title or relegation seasons are treated as rare but plausible), `is_promoted` in both models, one μ per team-season row.

2.1 Model 1 - `table_static.stan` (Static Skill)

Each team has one time-invariant latent vector `skill[t]` (sum-to-zero across teams), mapped to the points scale by a global coefficient `beta_pts > 0`. Season-to-season variation enters through observed, lag-safe covariates (z-scored on training data) and the promoted flag.

Model equation (for observation n , team `team[n]`, season implicit in the covariates):

$$Pts_{s,t} \sim t_{\nu}(\mu_{s,t}, \sigma_{pts})$$
$$\mu_{s,t} = \alpha + \beta_{pts} skill_t + \beta_{sot} x_{s,t}^{sot} + \beta_{lag} x_{s,t}^{lag} + \beta_{fomr} x_{s,t}^{form} + \beta_{promoted} is_promoted_{s,t}$$

Interpretation: α is the league-average points baseline; $\beta_{pts} skill_t$ is club t ’s persistent quality relative to that baseline; the three $\beta \cdot x$ terms are process adjustments from recent performance, $\beta_{prom} is_promoted$ is a fixed promoted-club adjustment, `skill[t]` is constant across all seasons for the same club.

2.2 Model 2 - `table_hierarchical.stan` (Hierarchical Skill)

Model 2 does not include `sot_diff_pg`, `pts_lag1`, or `ppg_last10`. Instead, expected points combine:

- `team_skill[t]` - persistent latent quality of club t (points scale),
- `season_effect[s]` - a single shared shift for all clubs in season s (league-wide high- or low-scoring year),
- `is_promoted` = as in Model 1.

Model equation:

$$Pts_{s,t} \sim t_{\nu}(\mu_{s,t}, \sigma_{pts})$$
$$\mu_{s,t} = \alpha + team_skill_t + season_effect_s + \beta_{promoted} is_promoted_{s,t}$$

For forecasting a new season, we use posterior `team_skill[t]`, draw a fresh shared `season_effect` for that season, and apply `is_promoted`; remaining team-specific noise is in the Student-t predictive residual.

2.3 Differences and Justification

Model 1 asks whether observed pre-season signals explain points beyond a fixed club rating. Model 2 asks whether a simpler latent hierarchy (club + shared season) suffices without those covariates. Prior scales are centered near $\tau_{\text{team}} \approx 12$ and $\tau_{\text{season}} \approx 5$ points (order of magnitude for cross-club and cross-season spread).

2.4 Technical Description - Stan Data Required

The table lists every data {} block entry each Stan program expects. One row is one team-season observation. Model 2 needs the extra season index `season[n]` (and scalar `S`) because it estimates a league-wide `season_effect[s]`; Model 1 pools all seasons under a single static skill[t] and does not use a season index.

Backtest training: $N = 320$, $S = 16$, $T = 42$. Training through 2025/26 (e.g. LOO in Section 4): $N = 340$, $S = 17$.

Stan key	Model 1	Model 2	Description
N	✓	✓	Number of team-season observations
T	✓	✓	Number of distinct teams in training (42)
S	-	✓	Number of seasons (16 for backtest)
team[n]	✓	✓	Integer team index
season[n]	-	✓	Integer season index
pts[n]	✓	✓	End-of-season points (target)
sot_diff_pg[n]	✓	✓	Z-scored shots-on-target difference per game
pts_lag1[n]	✓	✓	Z-scored previous-season points
ppg_last10[n]	✓	✓	Z-scored last-10-games PPG
is_promoted[n]	✓	✓	Binary promoted flag
nu	✓	✓	Fixed Student-t degrees of freedom (equals to 5)

3 Priors

3.1 Prior Selection Rationale

Priors are chosen from football domain knowledge and model scale. This distinction matters: if we tuned priors against the very backtest we later report as a performance metric, that metric would no longer be a fair test of out-of-sample performance, since the priors would have already "seen" the answer indirectly. Instead, every prior below is justified by reasoning that could have been done before looking at any model output. Key anchors:

- Mid-table baseline - intercept centred near ~50–52 points.
- Promoted clubs - weak negative prior on `beta_promoted` (substantial uncertainty allowed).
- Z-scored covariates (Model 1) - symmetric, broad Normal priors on `beta_sot`, `beta_lag`, `beta_form`.
- `beta_lag` - not given an extremely tight prior; lagged points may overlap with latent skill[t], but the prior remains weakly informative ($\sigma = 1$).
- Student-t $\nu = 5$ - fixed; standard robust-regression choice for outlier seasons.
- Log-scale priors on `sigma_pts`, `tau_team`, `tau_season` - positive scales with better sampling geometry.

Prior predictive checks in `02_model1_static_team` and `03_model2_hierarchical_team` verify that priors allow plausible point totals before fitting.

3.2 Model 1 Priors

Parameter	Prior	Rationale
intercept	$N(52, 15)$	League-average points baseline; wide uncertainty
<code>beta_pts</code>	$N(10, 10)$, lower=0	Maps latent skill to points; sign fixed so high skill=better team
<code>beta_sot</code>	$N(0, 10)$	SoT differential (z-scored); broad
<code>beta_lag</code>	$N(0, 1)$	Previous-season points (z-scored); moderate spread
<code>beta_form</code>	$N(0, 10)$	Last-10 PPG (z-scored); broad
<code>beta_promoted</code>	$N(-8, 8)$	Promoted teams may be weaker; wide uncertainty
<code>log_sigma_pts</code>	$N(\log 15, 0.6)$	Residual scale on log scale (~15 pts order of magnitude)
<code>skill[t]</code>	<code>std_normal()</code> , sum-to-zero	Latent static team strength
<code>nu</code>	fixed at 5	Student-t df (set in Python)

A few priors deserve closer explanation.

- The intercept $\sim N(52, 15)$ represents the baseline point total for an average team when all standardized covariates are near zero, latent skill equals zero and `is_promoted` is zero. The mean of 52 is treated as a broad midpoint of the Premier League table based on domain knowledge, rather than an empirical estimate from the data. The standard deviation of 15 implies an approximate 95% prior interval of 52 ± 30 (roughly 22 to 82 points), which flexibly accommodates both exceptionally weak and strong baseline seasons for a typical club.
- The latent team quality, `skill` $\sim \text{std_normal}()$, is defined on a standard normal scale. Because it is constrained as a sum-to-zero vector, the mean skill across all teams is exactly zero. This constraint ensures that the intercept captures the baseline level of an average team, preventing the intercept and team-specific effects from shifting freely relative to each other.

- The $\text{beta_pts} \sim N(10, 10)$ parameter maps one unit of latent skill to actual league points. In Stan, this parameter is constrained with $\text{lower}=0$, meaning the prior functions as a wide, strictly positive distribution. The prior mean of 10 suggests that a one-unit increase in latent skill corresponds to approximately 10 points, while the large standard deviation permits a significantly smaller or larger impact. This positive constraint establishes the orientation of the latent variable, ensuring that higher skill unambiguously translates to a higher expected point total. This $\text{lower}=0$ constraint is mathematically necessary because skill is an unconstrained sum-to-zero vector; without fixing the sign of beta_pts , the model would suffer from a global sign-flip non-identifiability. Constraining $\text{beta_pts} > 0$ successfully breaks this symmetry.
- The parameters $\text{beta_sot} \sim N(0, 10)$ and $\text{beta_form} \sim N(0, 10)$ correspond to standardized covariates. A one-standard-deviation change in these features could reasonably alter the final point tally by several points, but the priors are centered at 0 to avoid forcing a specific direction or magnitude before seeing the data. The resulting 95% prior interval of -20 to +20 points is intentionally broad.
- In contrast, $\text{beta_lag} \sim N(0, 1)$ is more regularizing. While points from the previous season are predictive, Model 1 already incorporates a latent team skill parameter. A very large lag effect would therefore double-count the same underlying signal of long-term team quality. Since the pts_lag1 variable is standardized, this narrow prior keeps the direct lag effect conservative unless the data strongly dictate otherwise.
- The $\text{beta_promoted} \sim N(-8, 8)$ prior expresses a weak domain belief that newly promoted teams may initially be weaker than established Premier League clubs. The mean of -8 is a prior assumption, not a fitted value. The approximate 95% interval spans from -24 to +8 points, which still allows for a negligible penalty or even a positive promotion effect if supported by the data.
- The $\text{log_sigma_pts} \sim N(\log(15), 0.6)$ prior describes the residual uncertainty on a positive scale via $\text{sigma_pts} = \exp(\text{log_sigma_pts})$, establishing a median residual scale of 15 points. Working in log-space allows for an unconstrained Normal prior while guaranteeing positivity after exponentiation - a standard Stan idiom that improves sampling geometry compared to a half-Normal prior directly on sigma_pts . Unconstrained, the 95% interval corresponds to $15 * \exp(\pm 1.2)$, or roughly 4.5 to 50 points; however, hard constraints of 0.5 and 30 are enforced in Stan for numerical stability. This setup keeps the residual noise on a realistic footballing scale without overfitting the prior to the observed dataset.

3.3 Model 2 Priors

Parameter	Prior	Rationale
intercept	$N(50, 25)$	League baseline; wider than Model 1
beta_promoted	$N(-5, 15)$	Promoted-team adjustment; only coefficient (no beta_sot, beta_lag, beta_form, no beta_pts)
log_sigma_pts	$N(\log 15, 0.8)$, bounded	Residual scale on log scale; same centre as M1, slightly broader SD
log_tau_team	$N(\log 12, 0.8)$, bounded	Scale of persistent between-team differences on points scale
log_tau_season	$N(\log 5, 0.8)$, bounded	Scale of league-wide season shifts (shared by all clubs)
team_skill_z[t]	std_normal(), sum-to-zero	Non-centered team quality
season_effect_z[s]	std_normal(), sum-to-zero	Non-centered season effect (not per club)
nu	fixed at 2	Student-t df (set in Python)

A few priors and structural choices in Model 2 merit closer explanation.

- The intercept $\sim N(50, 25)$ represents the expected point total for an average team in an average season, assuming non-promoted status. The mean of 50 serves as a clean, round baseline on the league scale, sitting squarely in the middle of the logically possible 0–100 point range. The standard deviation of 25 yields an approximate 95% prior interval of 50 ± 50 (roughly 0 to 100 points), which is intentionally broader than any specific historical average to avoid over-constraining the baseline.
- The beta_promoted $\sim N(-5, 15)$ parameter expresses a weak domain belief regarding newly promoted clubs. The prior mean of -5 suggests that these teams may face a slight performance penalty, but the substantial standard deviation of 15 implies a broad 95% prior interval spanning roughly -35 to +25 points. Consequently, the prior does not enforce a penalty; it flexibly accommodates severe disadvantages, negligible differences, or even highly successful breakout seasons.
- The model utilizes a non-centered hierarchical parameterization to capture team and season effects. The latent team quality parameter, $\text{team_skill_z} \sim \text{std_normal}()$, represents each team's persistent deviation from the league average. Because it is constrained as a sum-to-zero vector, team effects are strictly relative: a positive value denotes an above-average club, while a negative value signifies a below-average one. The magnitude of these differences on the points scale is governed by $\log_tau_team \sim N(\log(12), 0.8)$, where $\text{team_skill} = \text{tau_team} * \text{team_skill_z}$. The prior median for tau_team is 12 points. While its unconstrained 95% interval is broad- $12 * \exp(\pm 1.6)$, or roughly 2.4 to 59 points - hard boundaries of 0.5 and 45 are enforced in Stan for numerical stability. This specification readily allows for wide gaps between the top and bottom of the table without generating absurdly distorted simulated leagues.
- Similarly, $\text{season_effect_z} \sim \text{std_normal}()$ captures common macroeconomic shocks that affect all teams simultaneously within a given season. This vector also satisfies a sum-to-zero constraint, ensuring that the global intercept remains the true baseline for an average season rather than being absorbed by seasonal shifts. Crucially, Model 2 does not assign each club its own seasonal deviation (i.e., there is no team-specific season effect); season_effect is a single shared shift for the entire Premier League in a given year (e.g., a globally high- or low-scoring season). Any team-specific surprises left unexplained are absorbed by the residual scale and the Student-t likelihood.
- The magnitude of these shared seasonal shifts is controlled by $\log_tau_season \sim N(\log(5), 0.8)$. The prior median is set to 5 points, which is deliberately smaller than the median for tau_team.

This reflects domain knowledge suggesting that long-term differences between individual clubs are typically larger than baseline fluctuations between entire seasons. Its unconstrained 95% interval spans 1 to 25 points (bounded at 0.25 and 25 in Stan), allowing for moderate seasonal effects without permitting them to dominate intrinsic team quality.

- The $\log_sigma_pts \sim N(\log(15), 0.8)$ prior describes the residual uncertainty of a single season after accounting for team strength, the global season effect, and promotion status. The prior median is 15 points. Unconstrained, the 95% interval ranges from 3 to 74 points, but hard bounds of 1 and 35 are applied in Stan. This acknowledges that football seasons can be highly stochastic, while preventing the residual noise scale from exploding.
- Parameters such as σ_pts , τ_team , and τ_season are modeled on the log scale because scale parameters must strictly be positive; this standard Stan idiom guarantees positivity upon exponentiation while optimizing sampling geometry. Finally, prior predictive checks are treated strictly as a sanity check to verify that the implied point distributions are physically and footballistically plausible; they are not utilized to actively tune or overfit the priors to the observed data distribution.

3.4 Prior Predictive Checks - Parameters

For both models we draw parameters from the priors (respecting Stan bounds) and compare implied scales to football-plausible ranges. Scale parameters (σ_pts ; in Model 2 also τ_team , τ_season) remain positive with reasonable spread and do not pile up at zero. $\beta_promoted$ is centred below zero in both models, but the prior still allows positive values - especially in Model 2 ($N(-5, 15)$), which is intentional weak information, not a hard penalty.

Parameter prior quantiles (5% / 50% / 95%):

Parameter	5th %ile	Median	95th %ile	Model
intercept	26.7	51.9	76.2	M1
intercept	8.7	50.0	90.8	M2
β_pts	1.9	11/9	27.5	M1 only
β_sot	-16.2	-0.2	16.8	M1 only
β_lag	-1.6	0.0	1.7	M1 only
β_form	-16.3	0.3	16.8	M1 only
$\beta_promoted$	-21.3	-7.8	+5.4	M1
$\beta_promoted$	-30.3	-5.2	+19.5	M2
σ_pts	5.5	13.7	26.9	M1
σ_pts	3.7	12.9	30.5	M2
τ_team	3.1	11.6	34.4	M2 only
τ_season	1.3	4.8	16.1	M2 only

3.5 Prior Predictive Checks - Measurements

The second prior predictive check simulates team-season points: draw parameters from the prior, push them through the full generative equation (including Student-t noise), and compare to observed training points. This matters because individually reasonable parameters can still combine into implausible outcomes.

Model 1: prior predictive mean 50.9 points; central 90% interval -3.7 to 104.6. The interval is deliberately wide and includes technically impossible negative values; we accept this because priors are weakly informative and the posterior is constrained by the likelihood once real points are observed.

Model 2: prior predictive mean 48.8 points; central 90% interval -20.4 to 118.4. This is similarly broad, not tighter than Model 1: team_skill, shared season_effect, and sigma_pts are all weakly informative before data.

Observed training data (both models): mean 52.4, range 12–100. Both prior predictive distributions cover the observed support - the main pass criterion. Some prior mass on negative points is expected and acceptable at this stage.

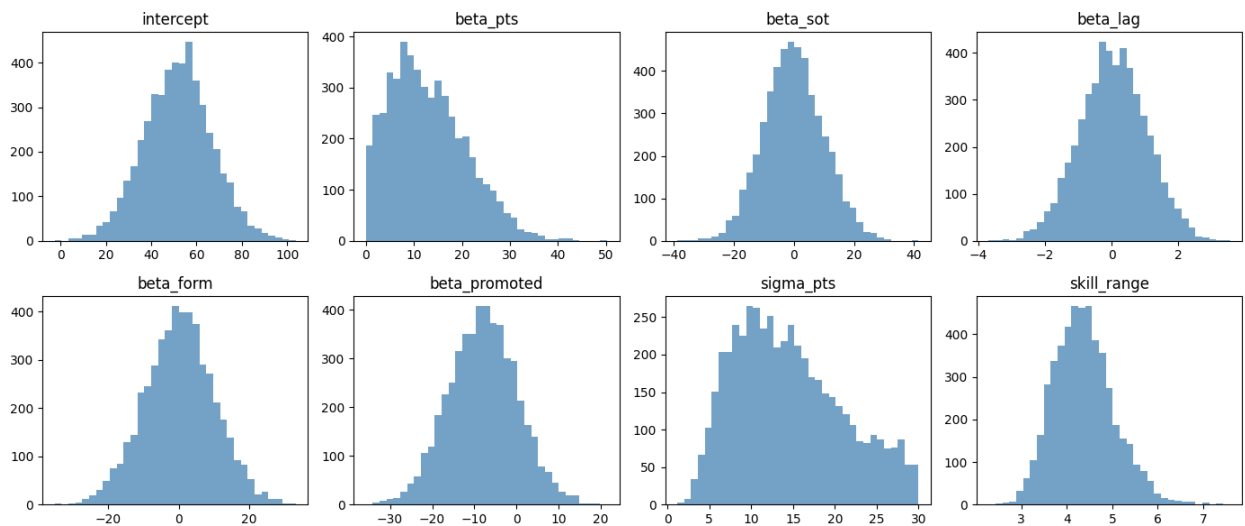


Figure 3.1 - Prior predictive parameter checks (Model 1): skill distribution and scale parameters.

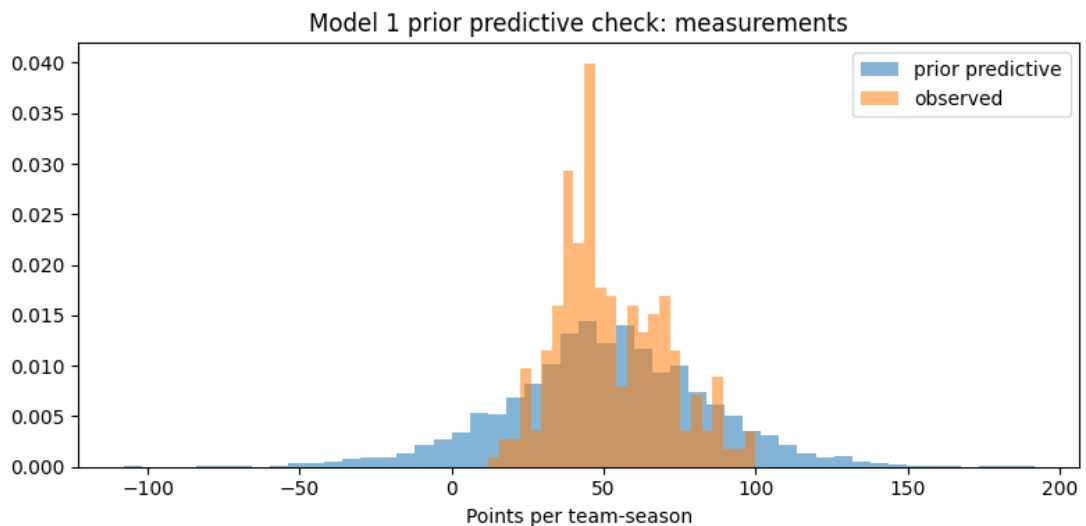


Figure 3.2 - Prior predictive measurement check (Model 1): simulated team-season points vs. observed range.

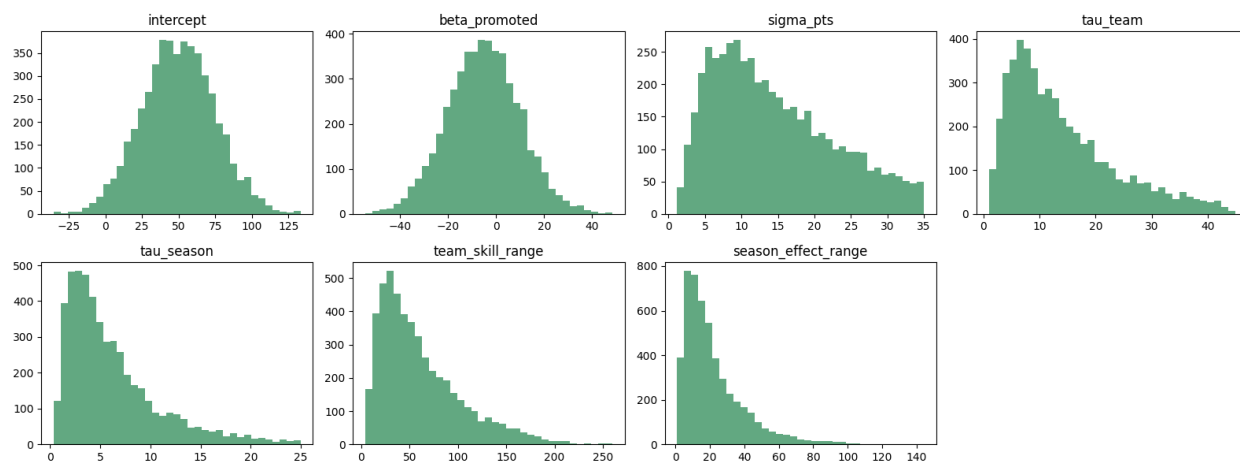


Figure 3.3 - Prior predictive parameter checks (Model 2)

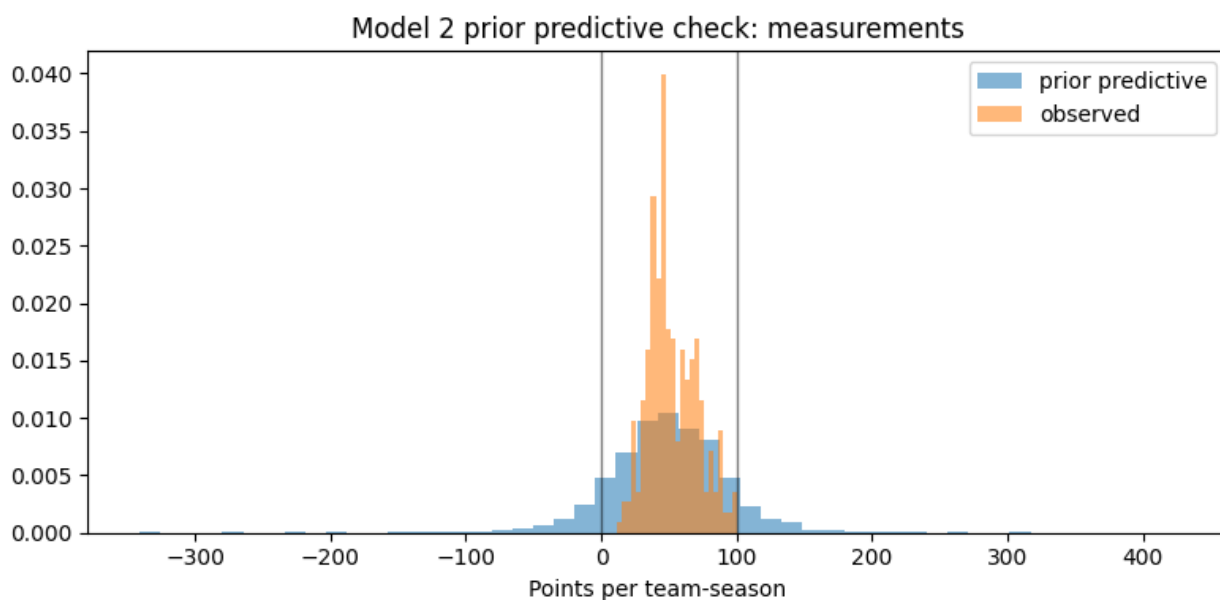


Figure 3.4 - Prior predictive measurement check (Model 2)

4. Posterior Analysis - Model 1 (Static)

4.1 Sampling Configuration and Issues

We fit Model 1 on the backtest training set ($N = 320$ team-seasons, seasons 2009/10–2024/25) using CmdStanPy with 4 chains $\times$ (1500 warmup + 1500 sampling), `adapt_delta = 0.99`, and `seed = 42`. `fit.diagnose()` reported no divergent transitions, satisfactory treedepth, E-BFMI, $\hat{R}$, and ESS. No parameters had $\hat{R} > 1.01$ or $\text{ESS}_{\text{bulk}} < 400$. No remedial steps were required.

4.2 Marginal Posterior Distributions

Table below reports marginal posteriors for the scalar parameters from the backtest fit: posterior mean, posterior standard deviation, and the 5th/95th percentile credible interval (CmdStanPy/ArviZ output). Convergence was satisfactory for all listed parameters (Section 3.6).

Posterior parameter summaries (backtest fit, $N = 320$ training rows):

Parameter	Mean	Std	5%	95%
intercept	50.18	1.08	48.24	52.27
beta_pts	7.63	1.45	5.10	10.54
beta_sot	6.27	1.04	4.23	8.13
beta_lag	0.03	0.86	-1.56	1.66
beta_form	1.20	1.09	-0.79	3.29
beta_promoted	-8.62	2.54	-13.24	-3/78
sigma_pts	8.72	0.47	7.84	9.68

Top team skills (posterior medians, Model 1):

Rank	Team	skill (mean)
1	Man City	2.42
2	Man United	1.82
3	Arsenal	1.71
4	Liverpool	1.47
5	Chelsea	1.41
6	Tottenham	1.37

This ranking broadly matches footballing intuition for the "big six" era of the Premier League, which is a useful, informal face-validity check - while we should not over-rely on subjective plausibility as a formal evaluation criterion, a model whose top-ranked clubs by latent skill bore no resemblance to the clubs widely regarded as historically strong would be a red flag worth investigating before trusting its forecasts.

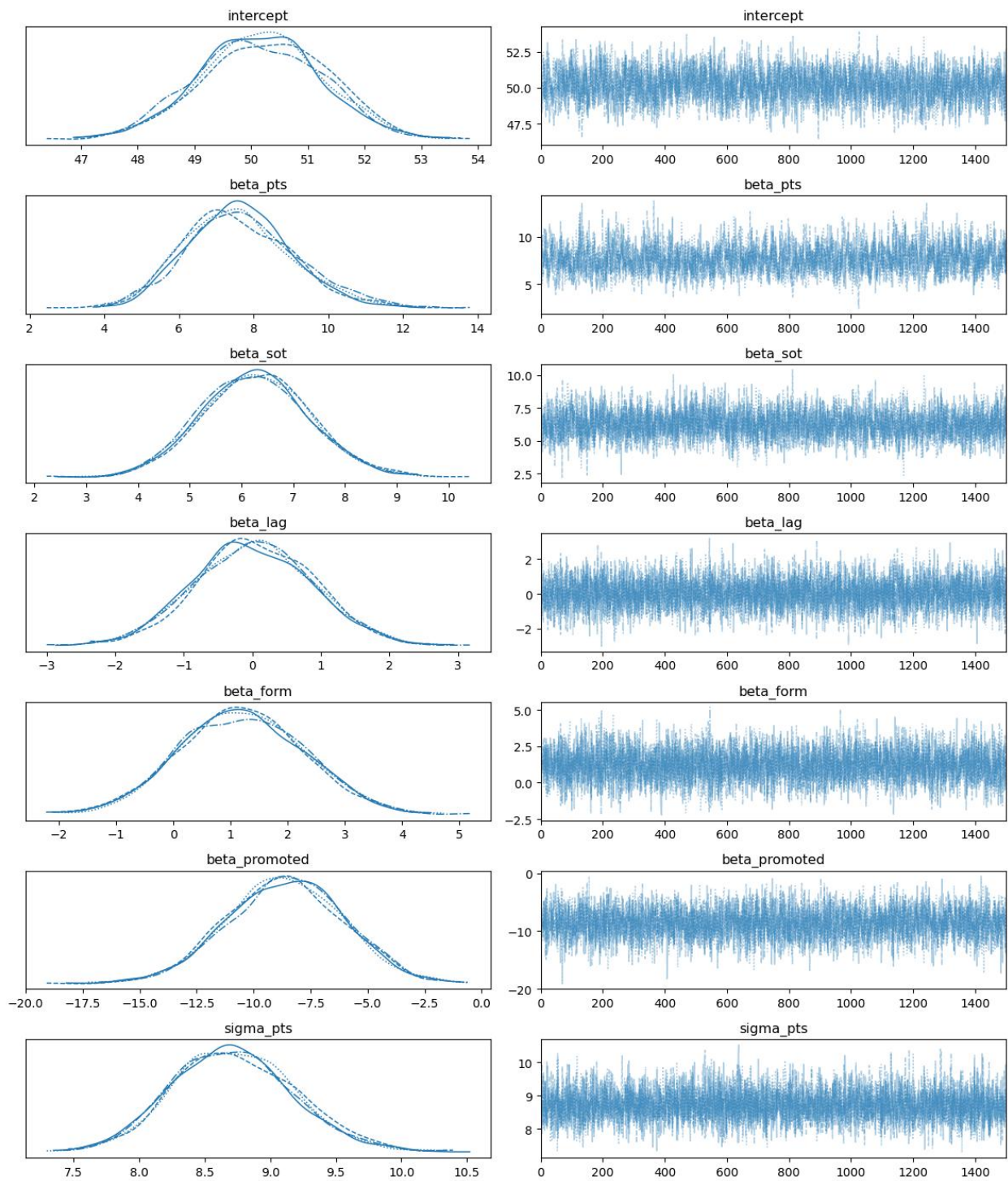


Figure 4.1 - Model 1 posterior marginal distributions: key parameters and team skill range.

4.3 Posterior Predictive Distribution

For each of the 6000 posterior draws (4 chains $\times$ 1500 post-warmup samples), we simulate a replicated training dataset of 320 team-season point totals. For each draw we compute the linear predictor from posterior parameters and observed covariates, then add Student-t noise ($\nu = 5$, matching the likelihood in `team_static.stan`). This is not done inside Stan's generated quantities block (which only stores `log_lik`).

We compare observed and replicated points on four test statistics (mean, standard deviation, maximum, minimum). For each statistic we report the Bayesian p-value: the proportion of replicated datasets whose statistic is at least as large as the observed value. Values near 0.5 indicate good calibration; values very close to 0 or 1 suggest the model struggles to reproduce that feature of the data.

Statistic	p-value	Interpretation
mean	0.38	Good - replicated means centred on observed
std	0.69	Good - model captures spread
max	0.85	Good - upper tail reproduced
min	0.08	Borderline - slight tension in lower tail

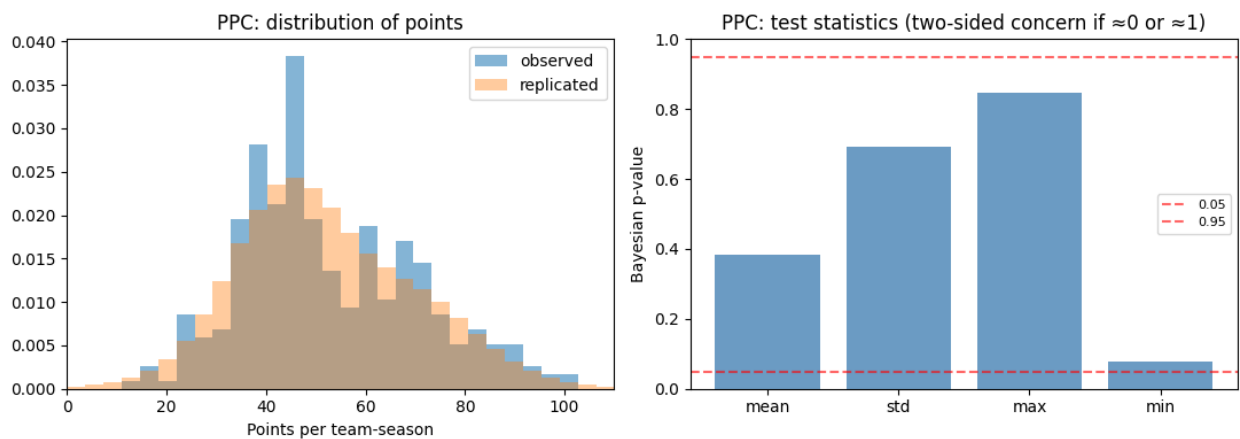


Figure 4.2 - Model 1 PPC: observed vs. replicated team-season points distribution.

4.4 Data Consistency

Observed and replicated point distributions align well. The minimum p-value (0.08) is borderline but acceptable for a course model. A value near 0 would prompt revisiting `sigma_pts` or adding covariates. No systematic over- or under-dispersion is evident.

4.5 Backtest 2025/26

Figure below breaks down the 2025/26 backtest by team: although average point error is modest (9.37), a few clubs account for most of the miss, while several mid-table sides are predicted almost exactly.

MAE points: 9.37

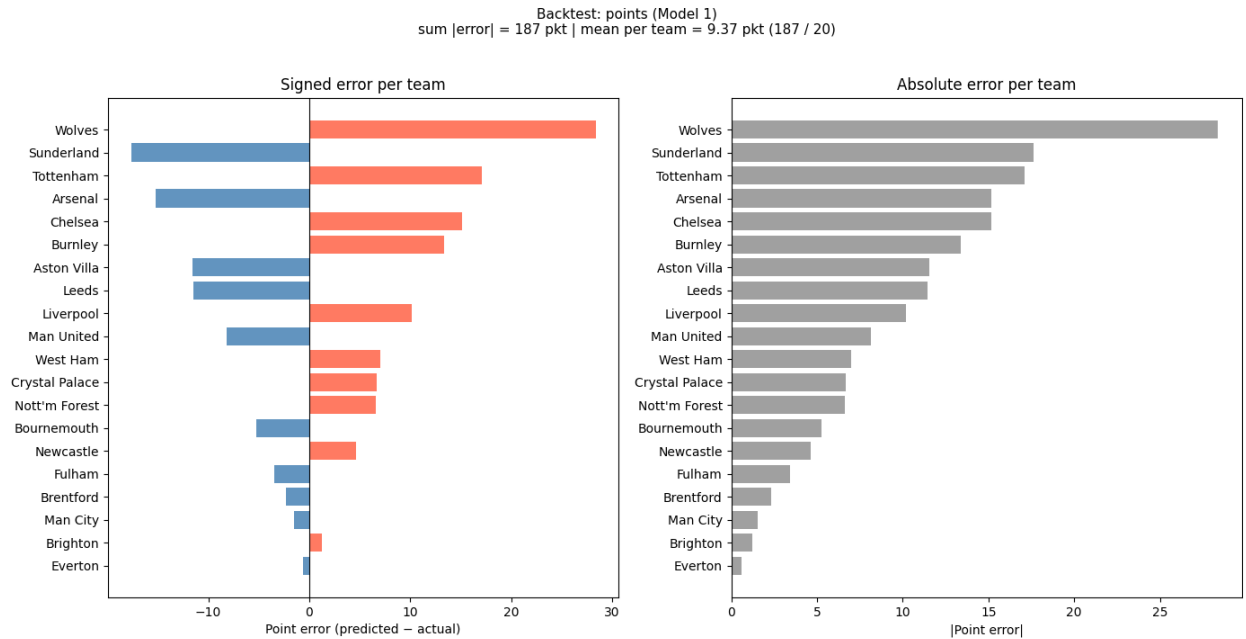


Figure 4.3 - Model 1 backtest 2025/26: Per-team point prediction errors.

MAE position: 3.80 (mean absolute position error across 20 teams).

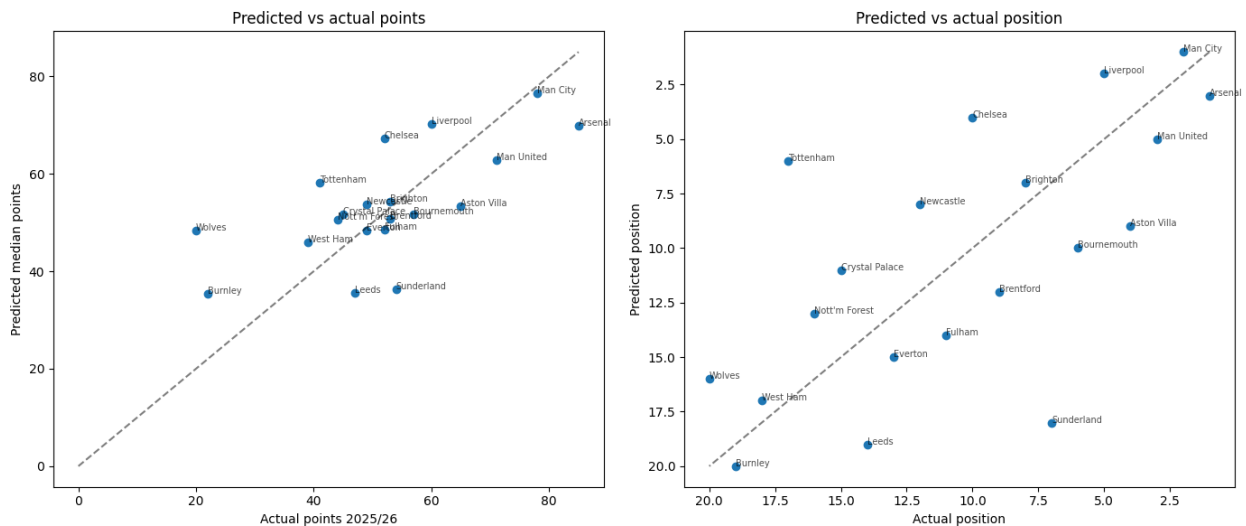


Figure 4.4 - Model 1 backtest 2025/26: predicted vs. actual league position (scatter).

5. Posterior Analysis - Model 2 (Hierarchical)

5.1 Sampling Configuration and Issues

We fit Model 2 on the same backtest training set ($N = 320$, seasons 2009/10–2024/25) with 4 chains $\times$ (1500 warmup + 1500 sampling), $\text{adapt_delta} = 0.99$, and $\text{seed} = 42$ - the same settings as Model 1. `fit.diagnose()` reported no divergent transitions; all $\hat{R} \leq 1.01$ and all $\text{ESS_bulk} \geq 400$.

Identifiability uses sum-to-zero standard-normal vectors: `team_skill_z` ($T = 42$ clubs) and `season_effect_z` ($S = 17$ seasons), scaled by `tau_team` and `tau_season`. This is a non-centered parameterisation (`team_skill = tau_team * team_skill_z`, `season_effect = tau_season * season_effect_z`). There is no per-team `season_dev[s,t]` in the current model; shared `season_effect[s]` captures league-wide season shifts, and team-specific surprises remain in the Student-t residual.

5.2 Marginal Posterior Distributions

Posterior parameter summaries (backtest fit, $N = 320$ training rows):

Parameter	Mean	Std	5%	95%
intercept	46.67	1.04	44.395	48.37
beta_promoted	-5.88	2.07	-9.31	-2.52
sigma_pts	8.01	0.47	6.46	8.01
tau_team	12.49	1.61	10.11	15.42
tau_season	1.21	0.52	0.52	2.20

`sigma_pts` is residual scale after `team_skill`, `season_effect`, and `beta_promoted`. Persistent between-club spread is mainly `tau_team` ≈ 12 points (top-to-bottom skill range ~ 49 points on the prior predictive check scale).

Top team skills (posterior medians, Model 2 - long-run component):

Rank	Team	team_skill (median, pts)
1	Man City	34.7
2	Arsenal	24.4
3	Chelsea	23.1
4	Man United	22.9
5	Liverpool	22.2
6	Tottenham	20.2

Unlike Model 1's latent skill scale, these are directly on the points scale. Man City leads clearly; Arsenal and Liverpool sit above Manchester United - a modest reordering vs Model 1's top-six ranking, but without per-club seasonal deviations we should not over-interpret small rank swaps.

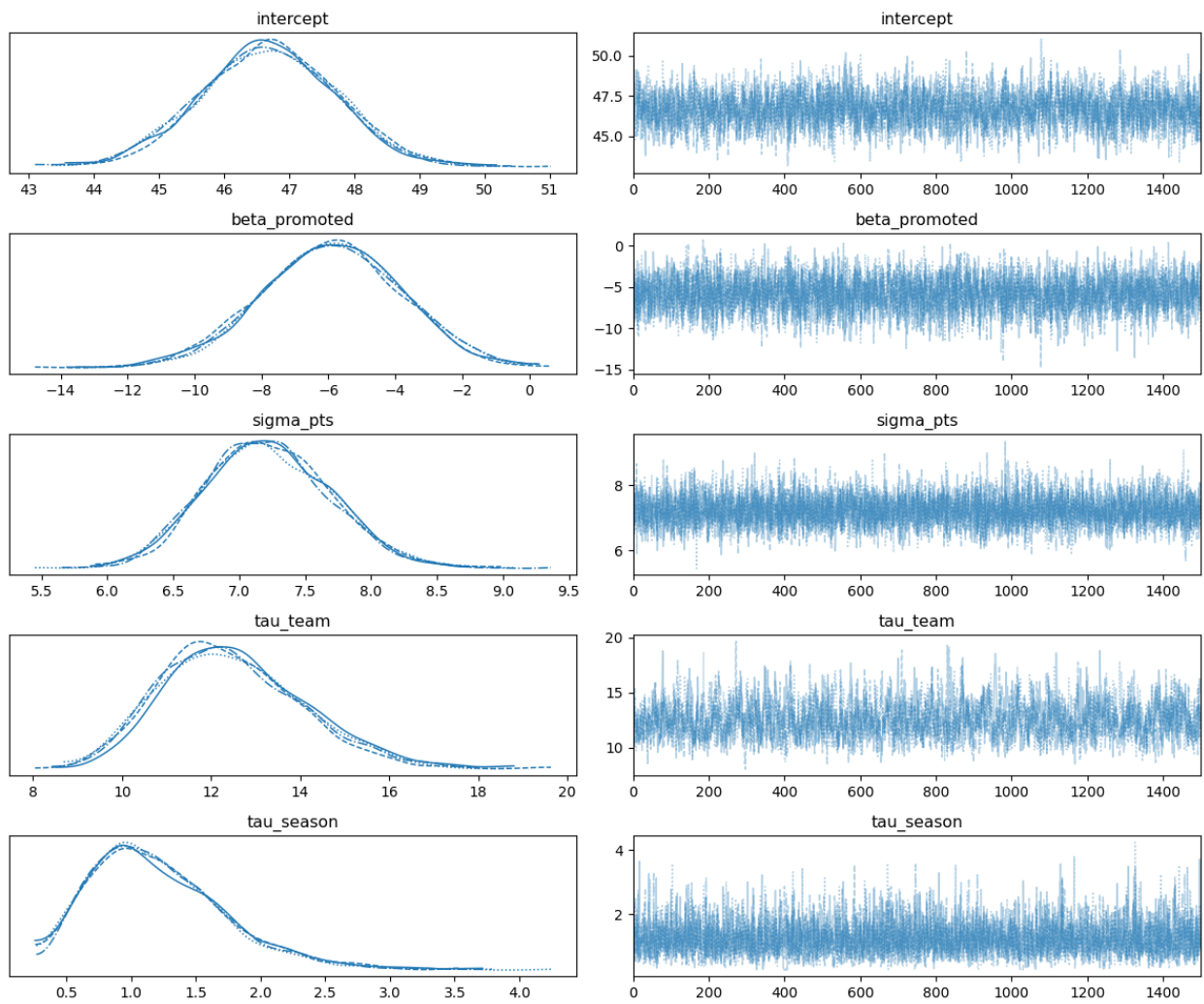


Figure 5.1 - Model 2 posterior marginal distributions: global parameters and hierarchical scales.

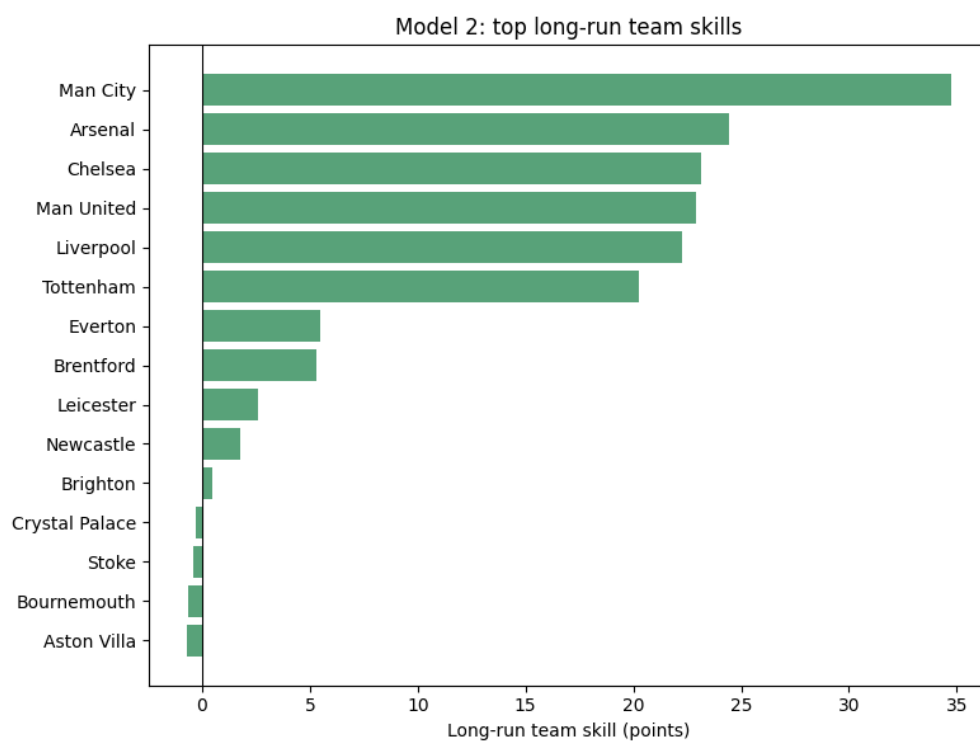


Figure 5.2- Teams skill over season.

5.3 Posterior Predictive Distribution

Statistic	p-value	Interpretation
mean	0.25	Good
std	0.19	Good
max	0.57	Good
min	0.18	Borderline (at the 0.05 threshold)

Compared with Model 1: mean and max are similar; std is essentially unchanged (0.19 vs 0.69). The minimum statistics are not better here - $p = 0.18$ vs 0.08 in Model 1 - so Model 2 does not clearly fix lower-tail tension on this single holdout training check.

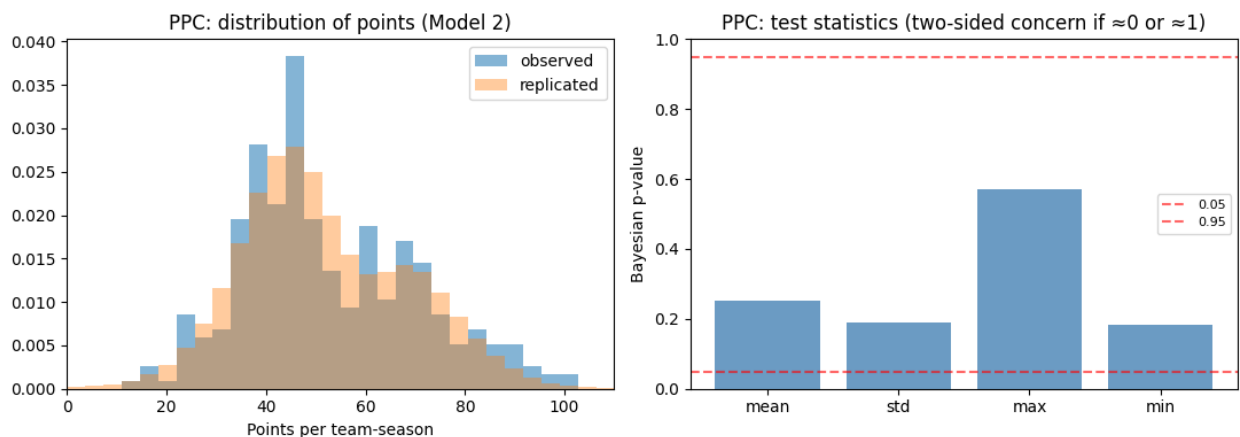


Figure 5.3 - Model 2 PPC: observed vs. replicated team-season points. Minimum p-value (0.20) is healthier than Model 1.

5.4 Data Consistency

PPC histograms overlap the observed distribution across most of the range; no p-value is extreme. σ_{pts} is similar to Model 1 (7.20 vs 8.72), not lower: extra structure is absorbed mainly by τ_{team} , not a much smaller residual. Overall, the hierarchical model is adequate for a course project, with the same mild lower-tail caveat as Model 1.

5.5 Backtest 2025/26

Same protocol as Model 1: train 2009/10–2024/25, predict 2025/26 (20 teams). Model 2 is slightly worse on this single holdout season, despite comparable PPC behaviour. That is a useful reminder: better hierarchical flexibility on training PPC does not guarantee better one-season forecasting - formal comparison (e.g. LOO) and more holdout seasons would be needed for stronger conclusion.

MAE points: 9.93

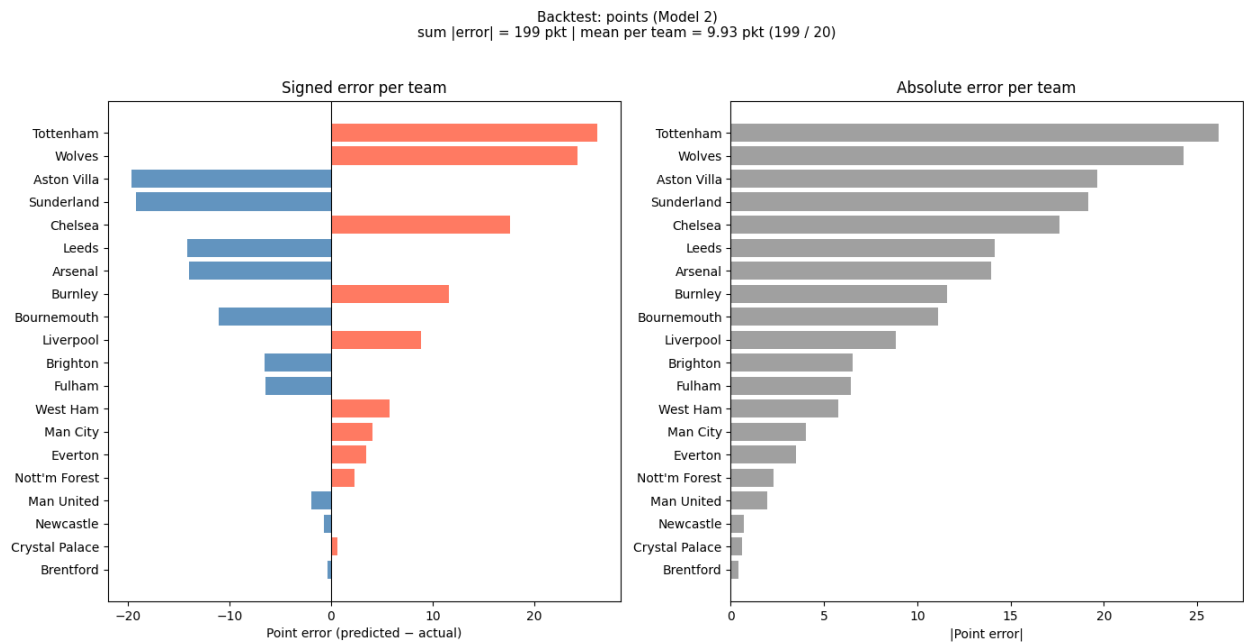


Figure 4.3 - Model 1 backtest 2025/26: Per-team point prediction errors.

MAE position: 4.10

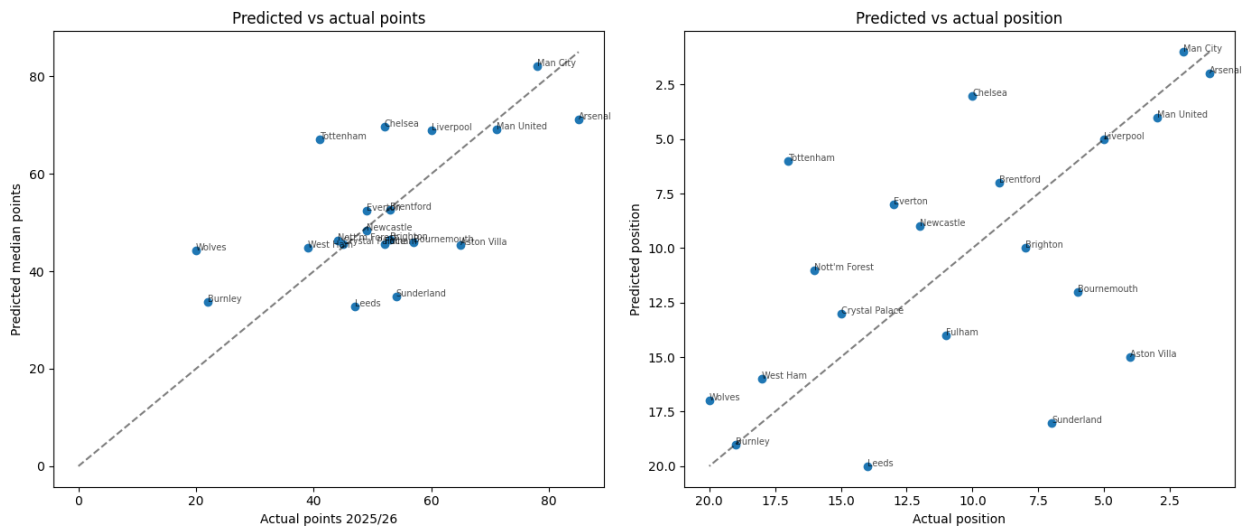


Figure 4.4 - Model 2 backtest 2025/26: predicted vs. actual league position (scatter)

6. Model Comparison

After fitting and validating both models individually, we ask which model to use and how confident we can be. We use two complementary angles: formal information criteria (PSIS-LOO, WAIC) on the full training sample, and a single-season holdout backtest (2025/26; Sections 3.5 and 4.5). There is no multi-season rolling backtest in this project.

6.1 Information Criteria (forecast training, N = 340 team-season rows)

Both PSIS-LOO and WAIC are computed on the full training set through 2025/26.

Criterion	Model 1 (Static)	Model 2 (Hierarchical)	Δ (M2 - M1)	Winner
elpd_loo	-1302.61 (SE 14.09)	-1332.63 (SE 14.13)	-30.02	M1
p_loo	37.4	51.55	-	-
elpd_waic	-1302.41 (SE 14.11)	-1332.25 (SE 14.14)	-29.84	M1
LOO weight	0.92	0.07	-	M1
Pareto k > 0.7	0 / 340	0 / 340	-	Both reliable
WAIC warning	Yes	Yes	-	Use LOO

6.2 WAIC Discussion

Both models trigger the ArviZ WAIC warning (posterior variance of pointwise log predictive densities exceeds 0.4 for some observations), so WAIC may be less stable. Model 2 has lower elpd_waic and higher p_waic (more flexibility), but still underperforms Model 1 on this metric. Given the warnings, PSIS-LOO is preferred over WAIC.

6.3 PSIS-LOO Discussion

All 340 team-seasons have Pareto $k \leq 0.7$ for both models, so LOO estimates are reliable. Model 1 is preferred (LOO weight 0.81 vs 0.19; $\Delta\text{elpd_loo} = -30.02$). This favours the simpler static model with process covariates over the hierarchical specification on in-sample predictive density - consistent with the 2025/26 backtest, where Model 1 also has lower MAE (points 9.37 vs 9.93; position 3.80 vs 4.10). We do not treat this as definitive proof that Model 1 is always better, but all three checks in this project point the same way.

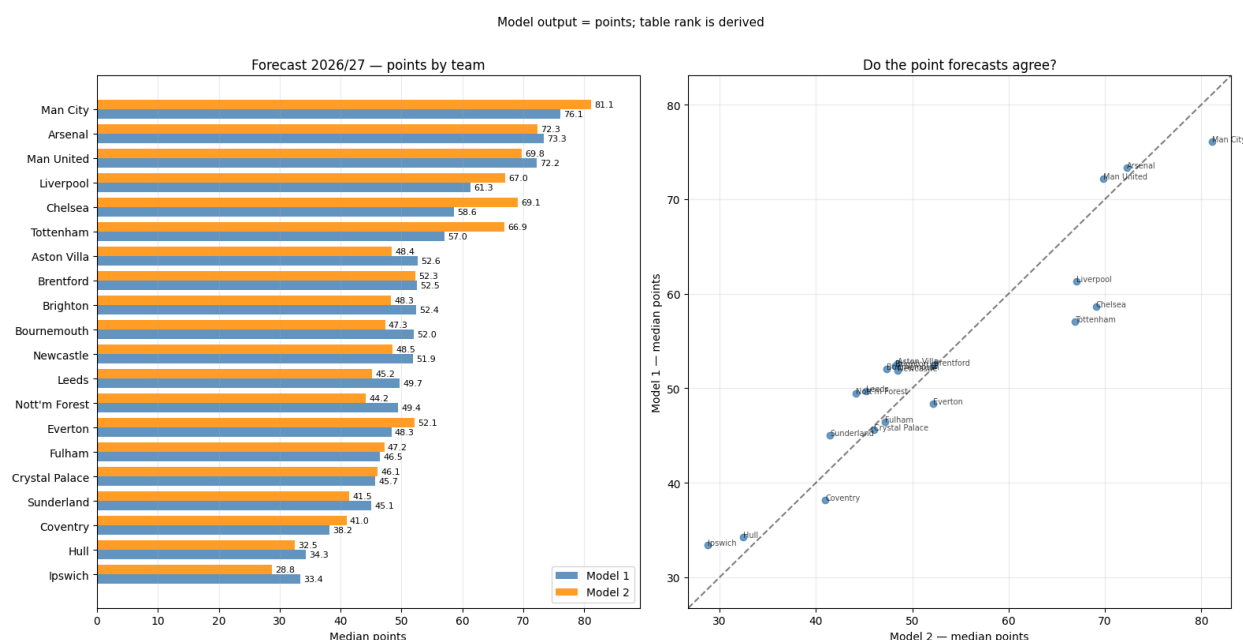


Figure 6.1 - Forecast 2026/27: Model 1 vs. Model 2 predicted positions (scatter) and points (column). Spearman cross-model correlation

6.4 Final Assessment

We compare models using PSIS-LOO/WAIC on full training ($N = 340$, through 2025/26) and a single holdout backtest (2025/26). There is no rolling backtest in this project.

PSIS-LOO is the preferred criterion given WAIC warnings (Section 5.2). Model 1 has higher `elpd_loo` (-1302.6 vs -1332.63; $\Delta = -30.02$; LOO weight 0.92 vs 0.07).

The 2025/26 backtest points the same way: Model 1 has lower MAE on points (9.37 vs 9.93) and positions (3.80 vs 4.10).

For 2026/27, the two models produce similar league orderings (same top three: Man City, Arsenal, Man United), but point medians differ more - especially for Chelsea, Liverpool, Tottenham, and Everton, where Model 2 predicts substantially more points. Position changes are mostly in the middle of the table (e.g. Everton 14 vs 8; Aston Villa 7 vs 10).

Recommendation: report both forecasts when they differ materially. Model 1 is the stronger model on in-sample LOO and the only holdout backtest we have. Model 2 remains a useful structural alternative (latent `team_skill` + shared `season_effect`), but we do not prefer it on predictive evidence in this project.

6.5 2026/27 Forecast - Side-by-Side

Both models trained on 17 seasons (2009/10–2025/26). Promoted: Coventry, Ipswich, Hull (is_promoted = 1). Model 1 also uses training-mean process covariates for new clubs; Model 2 uses promoted status only.

index	team	pos_m1	pts_m1	pos_m2	pts_m2	pts_diff	pos_diff
0	Man City	1	76.10	1	81.08	-4.98	0
1	Arsenal	2	73.32	2	72.27	1.05	0
2	Man United	3	72.16	3	69.79	2.37	0
3	Liverpool	4	61.31	5	67.05	-5.74	-1
4	Chelsea	5	58.64	4	69.06	-10.42	1
5	Tottenham	6	57.03	6	66.90	-9.87	0
6	Aston Villa	7	52.64	10	48.44	4.19	-3
7	Brentford	8	52.48	7	52.27	0.22	1
8	Brighton	9	52.38	11	48.26	4.12	-2
9	Bournemouth	10	52.04	12	47.31	4.73	-2
10	Newcastle	11	51.88	9	48.51	3.38	2
11	Leeds	12	49.68	15	45.19	4.50	-3
12	Nott'm Forest	13	49.42	16	44.18	5.24	-3
13	Everton	14	48.35	8	52.15	-3.80	6
14	Fulham	15	46.46	13	47.18	-0.72	2
15	Crystal Palace	16	45.65	14	46.06	-0.40	2
16	Sunderland	17	45.06	17	41.46	3.60	0
17	Coventry	18	38.18	18	40.98	-2.80	0
18	Hull	19	34.28	19	32.48	1.81	0
19	Ipswich	20	33.39	20	28.77	4.62	0

The top three are stable across models. Promoted clubs occupy positions 18–20 under both models, consistent with negative beta_promoted. The largest positional disagreement is Everton (7 places), not Arsenal. Point totals can differ by ~5–11 for several top-half clubs even when positions match, reflecting different model structure (covariates vs latent hierarchy). Mid-table teams with similar point totals (e.g. Brentford, Brighton, Bournemouth) illustrate irreducible ordering uncertainty

7 Notebook Index and References

Notebooks

- **00_project_overview_and_priors.ipynb** - project overview, model summaries, prior design.
- **01_eda_and_tables.ipynb** - EDA, team-season tables, promoted/relegated teams, data paths.
- **02_model1_static_team.ipynb** - Model 1: Stan fit, prior/posterior predictive checks, 2025/26 backtest.
- **03_model2_hierarchical_team.ipynb** - Model 2: Stan fit, PPC, latent team_skill summary, 2025/26 backtest.
- **04_forecast_2627_comparison.ipynb** - full-training refit (through 2025/26), PSIS-LOO/WAIC comparison, 2026/27 forecast side-by-side.
- **05_backtest_models_comparison.ipynb** - direct Model 1 vs Model 2 comparison on the single holdout season 2025/26 (not rolling).

Stan models

- **stan/team_static.stan** - static latent skill[team] + process/lag covariates → team-season points (Student-t).
- **stan/team_hierarchical.stan** - long-run team_skill[team] + shared season_effect[season] + is_promoted

References

- Data: [datahub.io — English Premier League](https://datahub.io/English-Premier-League) (originally football-data.co.uk)
- Stan: <https://mc-stan.org>
- CmdStanPy: <https://cmdstanpy.readthedocs.io>
- ArviZ LOO/WAIC: Vehtari, Gelman, Gabry (2017). *Practical Bayesian model evaluation using leave-one-out cross-validation and WAIC*. Statistics and Computing.
- BDA3: Gelman et al., *Bayesian Data Analysis*, 3rd edition.
- Dixon & Coles (1997). *Modelling association football scores*. Applied Statistics 46(2), 265–280. (Contextual; this project models team-season points, not match scores.)